frontiers in Education

TYPE Conceptual Analysis
PUBLISHED 25 September 2026
DOI 10.3389/feduc.2026.1860079

Check for updates

OPEN ACCESS

EDITED BY
Katrine Ellemose Lindvig,
University of Copenhagen, Denmark

REVIEWED BY
Jan Van Der Veen,
Eindhoven University of Technology, Netherlands
Olivier Morizot,
Aix-Marseille Université, France

*CORRESPONDENCE
Casper Feilberg
feilberg@ikk.aau.dk

RECEIVED 19 April 2026
REVISED 19 July 2026
ACCEPTED 27 July 2026
PUBLISHED 25 September 2026

CITATION
Feilberg C and Jæger K (2026) Qualifying the debate on interdisciplinarity considering 50 years of educational experimentation at Aalborg University. Front. Educ. 11:1860079. doi: 10.3389/feduc.2026.1860079

COPYRIGHT
© 2026 Feilberg and Jæger. This is an open-access article distributed under the terms of the Creative Commons Attribution License (CC BY). The use, distribution or reproduction in other forums is permitted, provided the original author(s) and the copyright owner(s) are credited and that the original publication in this journal is cited, in accordance with accepted academic practice. No use, distribution or reproduction is permitted which does not comply with these terms.

Qualifying the debate on interdisciplinarity considering 50 years of educational experimentation at Aalborg University

Casper Feilberg* and Kirsten Jæger

Department of Culture and Communication, Aalborg University, Aalborg, Denmark

Considering 50 years of pedagogical experimentation with problem- and project-based learning at Aalborg University, this paper investigates how interdisciplinarity as an educational principle has been realized in the university's teaching and learning practices. In order to analyze the initial approaches and experiments, we introduce two theoretical positions on disciplinarity and interdisciplinarity in education: the powerful knowledge approach, defending the value of disciplinary knowledge, and the German Didaktik and *Bildung* approach, requiring students to engage in boundary-crossing explorations of foreign knowledge. These approaches are used to interpret the initial experimentation with project work. At a policy level, the paper identifies an overall trend towards higher discipline-orientation in program offerings and curricular structures. Against this backdrop, the core parts of the paper discuss the pedagogical distinctions underpinning problem-oriented project work as it was practiced in the university's early years in order to identify and formulate more precisely the key qualities of Interdisciplinary Problem-oriented Project pedagogy (IPP) as it is ideally practiced today. By reinterpreting the work of Rousseau, Gadamer, and Negt from a Danish *Bildung* position (cf. Kemp and Nielsen), we shed new light on the pedagogical and curricular concepts that underpin the practice of IPP and its potential as a learning methodology for developing interdisciplinarity within human and social science degree programs. We argue that well-regulated freedom within the curriculum, termed a wide thematic framework, and student direction are the most powerful ways of inviting both interdisciplinarity and *Bildung* processes into problem-oriented project work. We find that in order for IPP to succeed, project work must be guided by the principle of exemplarity, supported by a *Bildung* culture within students, supervisors, and educational organizers, and the supervisor must be able to support the potential space, mimesis, and reversibility between symmetrical and asymmetrical roles along the process of supervision.

KEYWORDS
***Bildung*, exemplarity, interdisciplinarity, Negt, problem- and project-based learning, project pedagogy, project work, student direction**

1 Introduction

This paper will take its point of departure in the growing interest in interdisciplinarity in higher education (Lindvig, 2022; Lyall et al., 2015). Often incentives to include more interdisciplinary study activities come from "above", from university leaderships interested in demonstrating innovation and social responsibility. At the level of program coordinators and teachers, responsible for curriculum decisions, introducing (more) interdisciplinarity may be experienced as less straightforward. Many, if not most, university programs include both disciplinary and interdisciplinary elements, and faculty discussions around the balance between disciplinary depth and interdisciplinary openness/flexibility are inevitable, contributing to turning interdisciplinarity into a "hot topic" in HE curricula and to observable discrepancies between interdisciplinary intentions and realities (Lindvig and Ulriksen, 2019). As noted by Lindvig (2024), the age and history of institutions influence their approaches to interdisciplinarity: institutions with a long tradition of organizing themselves along disciplinary lines face challenges that differ from the ones experienced by "young" universities founded in the 20th century. However, as demonstrated by this paper's case research, this does not imply that young universities and their faculties do not struggle to find good models and practices for combining and balancing disciplinary and interdisciplinary elements in their curricula.

Consequently, this paper takes Aalborg University (AAU), founded in 1974, as context for a descriptive case study (Flyvbjerg, 2006; Gerring and Cojocaru, 2016). AAU – with its problem- and project-based approach - has grappled with the disciplinarity/interdisciplinarity balance for more than 50 years. Informed by university policies, curricula, current and historical descriptions of educational practices, this paper takes inspiration in the "insider action approach" (IRA), a form of qualitative research undertaken by organizational members investigating changes in their own organization (Coghlan and Holian, 2021). In this endeavor, we draw on documents (university policies, curricula, and descriptions of practice) as well as collaborative relations and deep contextual knowledge of the university's intellectual and cultural roots. The problem- and project-based approach implies that all curricula consist of a combination of discipline-specific content, often taught in courses, and project modules that to varying degrees open to inquiry across disciplinary boundaries as deemed relevant in relation to the addressed problem (Feilberg, 2022b; Jensen et al., 2019). Inspired by the student movement's radical critique of the discipline-organized university in the 1960's, the problem-oriented and project-based model initially abolished discipline-focused course teaching, particularly at Roskilde University (Whitehead, 1981). Disheartened by political reactions to such radical reforms, Aalborg University developed a more cautious approach and included disciplinary course teaching in its pedagogical model consisting of 50% course modules and 50% project modules, which to the present day dominates curriculum organization at AAU (Feilberg, 2022a, p. 42; Servant-Miklos and Spliid, 2017, p. 791). Still, however, concrete decisions on the role of the disciplines and of interdisciplinarity in curricula often rest on teachers' research and teaching tradition, political interests, and considerations of stakeholder satisfaction (students, censors, future employers). In this paper, we deliver a conceptual and principled contribution to thinking about interdisciplinarity and the disciplines in university curricula based on pedagogical reflections. The paper itself takes an interdisciplinary approach to understanding educational models and curricular developments. The two key concepts guiding the exploration of disciplinarity and interdisciplinarity in university curricula – "powerful knowledge" and "*Bildung*" – have their roots in educational sociology and educational philosophy respectively. However, investigating changes in educational practices throughout a long time period necessarily involves consideration of historical context and policy rationales as well as a deeper understanding of pedagogical implications of shifting balances between disciplinarity and interdisciplinarity. This investigation will proceed in two steps: firstly, part one) a conceptual discussion between the above mentioned approaches to thinking about disciplinary and interdisciplinary knowledge, and secondly part two) an inquiry into how disciplines and interdisciplinarity are balanced and supported in curricula and pedagogical practice within a problem- and project-based learning context.

2 Part one: initial analysis of curricular concepts supporting disciplinarity and interdisciplinarity

In this section, we present positions and distinctions for thinking about the roles of disciplinarity and interdisciplinarity in university curricula. Among the broad spectrum of approaches to understanding disciplinarity and interdisciplinarity in curricula, we select two approaches. While presented as distinct positions, we recognize that the positions also share important ideas such as, for example, the value of disciplinary knowledge. The two approaches also share an orientation towards contributing to socially just education systems – the powerful knowledge approach by emphasizing every student's entitlement to powerful disciplinary knowledge and the German Didaktik approach by considering student subjectivity and social change as integral elements of educational thinking. We will refer to the first approach as powerful knowledge theory, as contributors to this position (e.g., Bernstein, 1999; Muller and Young, 2019; Young, 2013), understand disciplinary knowledge as inherently empowering. Subsequently, we demonstrate how taking inspiration in the concept of *Bildung* originating in German philosophy and the German Didaktik tradition inspires curriculum thinkers, developers, and educators to open the curriculum to various forms of interdisciplinarity. The point of departure for turning to the *Bildung* concept is found in one of the relatively rare contributions that juxtapose the primarily Anglo-American curriculum tradition with the Didaktik tradition (Friesen, 2021).

2.1 The "powerful knowledge" approach

Contributors to the powerful knowledge theory strive to reinstate the importance of knowledge in the curriculum,

particularly disciplinary knowledge. It argues against a number of trends in curriculum studies, first and foremost instrumentalist approaches that seek to "optimize learning" and approaches that pay more attention to the social, cultural or gender origin of knowledge than to knowledge itself [often referred to as standpoint theory criticizing curricula for exclusively representing the "knowledge of the powerful" (Muller and Young, 2019)]. Positively, the powerful knowledge approach formulates three essential principles that may serve as inspiration for policymakers, curriculum developers, and teachers. Firstly, *knowledge in itself can be empowering.* For knowledge to be empowering, it must be specialized and differentiated from the student's lifeworld (Young, 2013). The capacity of knowledge to be empowering does not depend on the social origin of the knowledge, but on the quality of knowledge itself. Through its empowering effect, it offers the knower the possibility of drawing on its explanatory power, its applicability to a range of contexts, and its potential for discovery of the "not yet thought" (Bernstein, cited in Muller and Young, 2019).

Secondly, *powerful knowledge is, first and foremost, disciplinary knowledge*. Disciplinary knowledge holds more value, is simply "better" than other forms of curricular content. This argument is founded on a sociological understanding of the role of knowledge in human societies, particularly Durkheimian distinctions between "sacred" and "profane" knowledge (Muller, 2000; Young, 2013). Building on Durkheim's insight that scientific disciplines represent the sacred knowledge of modern society because this knowledge allows the knower to transcend – in thought – context-bound everyday experience, Young and colleagues argue that only disciplinary knowledge provides the knower with the transformative power of thinking beyond the already known. As argued by Muller (2000) "A position such as that of Durkheim is invariably in favor of disciplinary specialization" (Muller, 2000, p. 109). Thirdly, *every student is entitled to (access to) powerful knowledge* (Young, 2013).

2.2 The German didaktik and *Bildung* tradition

While powerful knowledge theory urges curriculum theorists (and practitioners) to reinstate knowledge as the basis of curricular thinking by asking them to focus on the knowledge to which students are entitled, critical voices counter that in the quest for stronger knowledge orientation curriculum thinking has gravitated too much towards knowledge *per se* at the cost of a sensitivity toward student subjectivity. The educational tradition that most radically places student subjectivity and fundamental change in the student's relationship to the world at its center is the *Bildung tradition,* inspired by, among others, Gadamer and Klafki (Friesen, 2021). According to Friesen, thinkers in the *Bildung* tradition introduce a terminology describing the change brought about by the type of engagement with knowledge that results in *Bildung*. In the following we will distill from contributions to the German *Bildung* tradition a number of concepts that demonstrate how a *Bildung*-inspired curriculum will need to move beyond the focus on disciplinarity prescribed by powerful knowledge theorists.

Three insights on disciplinarity and interdisciplinarity gained from engaging the *Bildung* tradition are presented below. Furthermore, key differences between the two approaches are highlighted in Table 1.

Firstly, we encounter in the *Bildung* literature a terminology for analyzing the relationship between student subjectivity and disciplinary learning. The encounter with (new) advanced and specialized knowledge generates an experience of entering a foreign universe, an experience of alienation as Gadamer (1960/2010) would phrase it. For the alienating experience to contribute to *Bildung*, the student must, however, come to feel at home in the disciplinary universe that was once foreign and return to oneself as changed. Obviously different types of experience (travelling, internships, relationships etc.) may have alienating and identity-changing effects. It is noticeable, however, that knowledge in itself represents *Bildung* opportunities, and that Gadamer's example of a *Bildung*-promoting activity (drawn from Hegel) is the reading of the classics, including learning the grammar of classic languages (Gadamer, 1960/2010; Sørensen, 2015). A *Bildung*-inspired curriculum offers opportunities for an alienating engagement of advanced knowledge and for students to make this knowledge their own in order to return to themselves as changed. As Feilberg (2014) explains, the necessary alienation involves "being one with the foreign in the sense of losing oneself in it [for examples by adopting new words and phrases and using them as if they were one's own (…)]" (2014, p. 101, our transl.).

Secondly, a *Bildung* inspired curriculum offers more than inclusion in a discipline and the opportunity to make disciplinary knowledge one's own. A "*sense for the time*",

TABLE 1 Comparison of key ideas of the powerful knowledge and the German Bildung approach to curriculum.

Powerful knowledge approach	German *Bildung* approach
Knowledge included in the curriculum should be specialized and differentiated from the student's lifeworld.	Knowledge included in the curriculum should enable the student's "alienation" and return (to oneself as changed).
What matters is the quality of knowledge. Quality knowledge is disciplinary knowledge.	What matters is the student-knowledge relation.
The mastery of disciplinary knowledge leads to empowerment.	Interdisciplinary learning leads to understanding one's own disciplinary perspective from the perspectives of other disciplines, "sensitivity towards the Other".
Education should provide opportunities of acquiring the disciplinary knowledge to which the student is entitled.	Education should provide opportunities of learning beyond one's discipline to obtain "a sense for one's time", insight into the defining problems of the historical moment in which one exists.

Disciplinarity and interdisciplinarity within the powerful knowledge and the German *Bildung* approach to curriculum – comparison of key ideas.

knowledge and understanding of the historical, cultural, and social conditions of the present, insight in epochal issues and problems, and the moral, political, and aesthetic claims that these insights raise towards individuals and communities, forms an important part of *Bildung*. Gadamer's (1960/2010) analysis of the concept of "taste" illustrates this point. Taste clearly refers to aesthetic judgment but rests on a foundation of moral, cultural, political, and historical knowledge. In his explanation of Klafki's *Bildung* concept, Sørensen refers to this sense as a "historically informed consciousness of crucial contemporary problems" (Sørensen, 2021). Such a sense is both individual and collective in nature: "*These are problems that we all, as citizens and human beings, each by ourselves and individually, confront to some extent and therefore must assume responsibility for, and this implies being willing to contribute to the solution of these problems*" (Sørensen, 2021, p. 16). This dimension of *Bildung* has contributed to a renewed interest in the concept, for example in relation to climate crisis education (Servant-Miklos, 2024; Slothuus, 2025). Ensuring that the curriculum supports this *Bildung* dimension can take many forms, ranging from inserting particular knowledge elements (e.g., sustainability knowledge) in the curriculum to supporting students' own explorations into epochal problems, as it happens in some versions of problem-oriented project work. Regardless of particular curricular realizations, enabling the development of a "sense for the time" will involve interdisciplinary learning.

Thirdly, *Bildung* includes an additional form of insight that requires students to cross disciplinary boundaries. Again, we can find the roots of this insight in Gadamer's *Bildung* concept: "*Empfänglichkeit für das Andere*" [sensitivity towards the Other], the ability to open up to different perspectives and, not least, to be able to see oneself and one's own perspectives as others may see them (Gadamer, 1960/2010; Grondin and Plant, 2003, p. 25). The individual must gain the ability to keep in mind the perspectives of others (Gadamer, 1960/2010, p. 23). To be able to observe one's own (for example disciplinary) perspectives from the outside, from the perspective of other disciplines is increasingly seen as important, not only as a personal capacity for empathy, but also as a professional skill enabling interdisciplinary and interprofessional collaboration around the solution of complex tasks. The capacity for self-reflexivity demanded by the *Bildung* concept has, however, an even broader scope. For Gadamer, to be a "gebildete person" does not only require the ability to observe one's own perspectives from others' positions, but also from broader societal (*allgemein*) perspectives (Gadamer, 1960/2010, p. 23). Like *Bildung,* the German word "*allgemein*" is difficult to translate. "Universal" or "general" signals vagueness or abstraction and does not fully cover the meaning of taking everyone, the collective, society as a whole into account. Nevertheless, *Bildung* asks the individual undergoing *Bildung* to see one's own (academic, professional, and social) existence as contingent upon and committed to broader societal, in other words, political perspectives.

The *Bildung* concept's call for elevation to "*Allgemeinheit*", the knowledge and understanding of societal and political conditions and opportunities for change has been further developed by contributors to Northern European educational thinking. Particularly Wolfgang Klafki and Oskar Negt translated the ambition of educating for political enlightenment and democracy into concrete didactic principles. Although scholars of the Critical Theory tradition have had their reservations about the *Bildung* concept because of its historical roots in bourgeois education institutions, they also developed *Bildung*'s potential for critical thinking and social change. In their work, Klafki and Negt (Negt, 1971; Feldt, 2023) laid the groundwork for didactic and curricular approaches that aim to cultivate students' ability to transcend the particular in the form of *teacher chosen* exemplary problems, critically interrogate their hidden structural conditions, and translate insights into meaningful social and communicative action. In the 1970s, the problem-oriented, project-based learning tradition, institutionalized in the educational philosophies of Denmark's two youngest universities (Roskilde University, RUC, and AAU) adopted key ideas from critical theorists' conceptualization of *Bildung*, first and foremost the concept of "exemplarity". Combined with the Danish tradition of group work (Christensen, 2020), experimentations with exemplary teaching and project work in schools and Higher Education took off. The concept of exemplarity was interpreted in several ways with respect to project work. According to the influential voices of Illeris (1974) and Hultengren (1976) exemplarity in project work involved that students should work on particular problems of such exemplary nature that they allow for in-depth acquisition of theoretical knowledge and insight into the structural conditions of the existence of the problem: "problems…should be representative of a larger and essential area of reality" (Illeris, 1991, p. 47). Both universities adopted the idea of an interdisciplinary, problem-oriented "basic education" (one year, or in RUC's case, two years) devoted to introductory studies across multiple disciplines within a broadly defined thematic frame. At AAU, the basic year constituted the first year of every student's curriculum and involved interdisciplinary student projects addressing broad thematic frameworks and cross-departmental cooperation between teachers, including allocation of supervisors across departmental boundaries. Subsequently, the basic year was shortened to one semester and eventually phased out.

Project work at AAU and RUC was often described as participant directed (Feilberg, 2022a, b), though it had different meanings. At RUC, participant direction meant that both supervisor and student's interests should be taken into consideration according to Illeris: "Participant direction […] indicates that the studies should be directed jointly by students, tutors, and other relevant participants. While all participants are equal, they have different functions and responsibilities" (Illeris, 1991, p. 46). Ideally, these research processes also involve a multifaceted engagement of reality enabling both practical-professional insights and theory-based knowledge. Such processes encourage transcending disciplinary boundaries to explore "foreign" knowledge resources and their potential for addressing the problem at hand.

Problem- and project-based learning represents a radical departure from a well-ordered induction into a specific discipline. Student subjectivity through participant direction takes center stage as groups of students and their supervisor define the question through which they seek, identify, and come to understand knowledge. Through these questions, students engage

disciplines as "complex, contested, and permeable" (McArthur, 2010, p. 306). In their research, students are inclined to move within and across disciplines. The interdisciplinarity of problem- and project-based learning emerges in the meeting of disciplines necessary for the investigation of a problem, in students' borrowing and integrating elements from different disciplinary knowledge bases in their work. Although not explicitly branded as "interdisciplinary pedagogy" (Slothuus, 2025), the problem- and project-based approach means that students routinely address the interdisciplinary nature of complex problems (Klein, 2009) and experience the potential and limitations of their own disciplines.

3 Part two: lessons learned from 50 years of pedagogical experimentation with interdisciplinarity at Aalborg University

Like many other universities, AAU has experimented with, and continues to explore, ways of supporting interdisciplinarity and student involvement. A culture that fostered interdisciplinary project work was especially developed in the basic year studies during AAU's early years. While rationales for abandoning the basic year differed across faculties and departments, a main reason for redefining the first year from an interdisciplinary introduction to university studies to an integral part of the disciplinary program was the need to provide a comprehensive introduction to each program's disciplinary content, among other things as a consequence of the Bologna Process, requiring a shortening of first cycle university education from four years to three years. As a consequence of ongoing structural changes, an ongoing challenge remains to articulate and theorize pedagogical distinctions that do justice to how interdisciplinary project work is currently practiced.

Thus, in this section we continue the discussion of interdisciplinarity, but shift the analytical focus from an international and general perspective on problem- and project-based learning towards a case analysis of the contextualized practice of problem-oriented project work at the Faculty of Social Sciences and Humanities at Aalborg University[1]. We present our interpretation of what we identify as current best practices and propose new pedagogical distinctions and models that can advance interdisciplinarity in education. We do this by discussing the work of Illeris (1974), Hultengren (1976) and their interpretation of Negt (1971), which played a formative role in the establishment of problem-oriented project work at AAU[2], as seen in relation to later developments within critical theory (cf. Negt, 1994; Nielsen, 1997), and perspectives from philosophy of education (Kemp, 2011).

While some hold the view that project work is based on the Northern European Didaktik tradition (Illeris, 1974), others locate it within the tradition of critical theory (Hultengren, 1976), and still others argue that it is grounded in constructivist learning principles (Dirckinck-Holmfeld, 2002). Based on the existential-phenomenological lifeworld ontology of Merleau-Ponty (2012), we distinguish between the phenomenon and lived practice, on the one hand, and the descriptions and theoretical terminology used to "point to" aspects of these practices, on the other (Feilberg, 2022a). This implies that pedagogical practices may be interpreted through different theoretical pre-understandings.

Our model, called Interdisciplinary, Problem-oriented Project pedagogy (IPP), is grounded theoretically in the tradition of hermeneutical and existential-phenomenological pedagogical philosophy (Feilberg, 2014, 2022a, 2019; Kemp, 2011; Thøgersen, 2010). A characteristic of any pedagogical tradition is the analysis of problems in society and the discussion of ethical and normative ideals. Educators must reflect on these ideals in order to form their own position and understanding of the paradoxes of education, e.g.,: while instruction and examination can ensure certain skills and competences in a student, the formation of an independent habitus of critical thinking and ethical values is something that only the students can bring about themselves, by embarking on *Bildung* processes (cf. Thøgersen, 2010, p. 51; Feilberg, 2019, p. 130). Accordingly, to reflect on power dynamics within education is important as pioneered by Rousseau (1762/1979), the founder of modern pedagogy. In the words of a Danish proverb, with respect to *Bildung*, you can bring the horse to the trough, but only the horse decides whether to drink or not. From this perspective, we can invite student subjectivity and autonomy into the educational activities as educators, but we cannot demand it or assume it out of respect for the individual student. On this point the Danish tradition differs from the German *Bildung* tradition, which does not thematize the freedom and individuality of the person, focusing instead on the collective tradition. Against the many power dynamics and stakeholders in a student's lifeworld - including parents, peers, teachers, and other "influencers" and forces - a student must struggle to identify a voice of their own, their individuality. We recognize Kemp's characterization of the Danish educational system as suffering from half-education (*Halbbildung*), meaning "the notion that narrows *Bildung* to knowledge of methods, competencies, and technical and/or scientific skills, while subordinating — or rendering irrelevant — the development of a sense of the individual's freedom and authority [myndighed] within the community, in favor of […] the logic of the market" (2015, p. 57, our transl.). Developing a sense of the individual's freedom and authority within the community is at the heart of IPP. While the curriculum organizer and supervisor of IPP can create project modules with a broad thematic framework in the curriculum - encouraging students to pursue their interests and opening up potential for interdisciplinary project work and *Bildung* processes - they must accept that students may choose differently.

[1]Our references of experience stems mostly from the study programs of international studies and psychology.

[2]As documented by Servant-Miklos (2020) and Feilberg (2022a,b), the tradition of problem-oriented project work at Roskilde University (RUC) and Aalborg University (AAU) in Denmark from the 1970s, and the tradition of problem-based learning (PBL) from McMaster University in Canada in the 1970s are fundamentally unconnected, and only the tradition at RUC and AAU has project work and interdisciplinary problem-orientation as core principles from its foundation.

3.1 Three key aspects of interdisciplinary project work (IPP)

In order to describe key qualities of Interdisciplinary, Problem-oriented Project pedagogy (IPP) within a study programme, we distinguish between three aspects of the model: A) *the epistemological aspect* concerning the scientific inquiry (cf. the research question) which the project group selects within the wide thematic framework of the project module (cf. the learning outcomes of the curriculum), B) *the interactional aspect* concerning the processes of the individual students and of the project group as a work group and the collaboration between the group, the supervisor and the external partners (cf. the learning outcomes of the curriculum), and C) *the pedagogical aspect* concerning the art of supporting IPP in practice as well as the theoretical concepts and ideals underpinning practicing project work as a student, a supervisor and an educational organizer.

An important prerequisite for IPP is the creation of a *pedagogical and interactional culture* among students, between supervisor and project group, and between supervisor and educational organizer that *holds* (cf. Winnicott) students and their project group so that they can carry out student-directed, interdisciplinary project work from the brainstorming phase, group formation and problem identification phase, to the oral group defence of the final academic report. When first semester students understand what problem-oriented project work entails, they are often both overwhelmed and intrigued to realise that *they formulate* the research questions, and not only respond to those of the teacher. The pedagogical model of IPP is only realisable through induction and the pedagogical and interactional culture. The latter is upheld through culture bearers and dialogues between supervisors with structural support from the study programme, management and the educational institution.

We start by discussing key pedagogical distinctions and later present qualities of IPP.

3.2 Exemplarity, interdisciplinarity project work and *Bildung* – then and now

During the first years at AAU, notably Hultengren developed and experimented with problem-oriented and interdisciplinary project work inspired by not least Negt's (1971) seminal work on sociological imagination and the principle of exemplarity, with great impact (Hultengren, 1976; Hultengren, 2014, p. 8). We will explore the original concept, its development, and its current meaning in IPP.

Disciplinarity and interdisciplinarity. The principle of exemplarity is of relevance to interdisciplinary education in several ways. First, Negt criticizes Wagenschein's original formulation of the principle for being held back by a concept of history limited to the dimension of the past, and a lock-in caused by "the traditional division of labour between the disciplines" (Negt, 1971, p. 26, our translation). Negt therefore finds inspiration in Mills's concept of sociological imagination, which according to Negt sets the exemplary learning free by building in the worker the ability to change between different perspectives and levels of investigation, from the political to the psychological, from the housekeeping of the family to the housekeeping of the state, in order "to discern the structural relations between an individual life history, immediate interests, desires, hopes, and historical events" (Negt, 1971, p. 28, our transl.). In Negt (1971), the principle of exemplarity is closely connected with the concept of sociological imagination (Nielsen, 1997, p. 286). In his interpretation of the principle of exemplarity, the ability to apply different perspectives was a necessary part of moving from the particular to broader perspectives, cf. sociological imagination (Nielsen, 1997). This is still very much the case at AAU. In current educational practice, exemplarity refers to both course modules and project modules: courses must teach "exemplary elements (theory, empirical knowledge, and methods) that represent the broader field", and within the project module students must "engage with the complexity of the problem as a whole and […] place the specific within a wider context" (AAU, 2025, p. 13).

What we term the exemplary breadth-and-depth model of IPP, denotes the order of first introducing students to two or more disciplinary courses that cover an exemplary selection of theoretical and/or methodological knowledge, and then offering a project module with a wide thematic framework for interdisciplinary project work, within the same semester or across two semesters. If students' problem identification and student direction are supported, it creates favorable conditions for the emergence of narrow interdisciplinarity in their project work. This can later expand into broader interdisciplinarity beyond the programme, provided they are taught philosophy of science along the way.

Exemplarity then and now. Who chooses the problems in the educational context of Negt (1971)? Negt never addressed a project work learning methodology himself, as the teaching he describes in his publications stems from classroom teaching activities and trade union education in Germany (*Arbeiterbildung*). The aim of Negt (1971) was to develop political *Bildung* within union workers' education. In Negt's (1971) accounts of exemplarity in the teaching of union workers, the *teacher* formulates the problem case while trying to include the interests and the conflicts experienced by the worker. The exemplary *Bildung* value (*Bildungswert*) of a teaching activity is - according to (Negt, 1971, p. 97) – dependent on three dimensions: 1) its proximity to individual interests, 2) the elements that extend beyond the immediate interests and engage with broader societal and structural relations, and 3) the significance that the educational content can have for the workers' emancipation. The aim was for the union worker to work from the example or experiences of the worker in a way in which the particular problem is understood in broader (*allgemein*) juridical, theoretical and societal perspectives (Negt, 1994, p. 48). These perspectives were rooted in critical theory, which pursues an emancipatory critique of exploitation and capitalism.

When Hultengren and others at AAU interpreted problem-oriented project work in the light of Negt's principle of exemplarity, student's participant direction in choosing the research question and problem was emphasized. But when looking back in 2014, to Hultengren this remained more of an

unrealizable ideal: "For many of us, it was an ideal notion that the students themselves could identify a problem they perceived as authentic, which the student group could then unfold into a deeper understanding of broader social relations. However, only very few of our students could, on the basis of their own lives, present a "problem" with the quality needed to function as exemplary material" (Hultengren, 2014, p. 8, our transl.). These observations resonate with publications from that period (Hultengren, 1979).

While Hultengren blames the problems with realizing the principle on the students (e.g., that they lack the experiences of the worker in a capitalist society), we argue that Hultengren and Negt actually put their emancipatory ambitions above the students' autonomy. As Negt argues: part of the critical potential of the principle of exemplarity stems from education and the creation of free spaces (*frirum*) set apart from everyday life and dominant pre-understandings, within which exemplary learning can unfold (Nielsen, 1997, p. 279). These free spaces and exemplary learning processes, argues Negt, break with the learners' immediate "closeness to their own interests and needs" so that they may become "conscious of the ways in which these are mediated and conditioned", and thus move toward a "reflexive development of broader societal relations derived from particular, pregnant points" (Negt, 1994 cf. Nielsen, 1997, p. 279, our transl.). We argue that Negt (1971) and Hultengren (1979) and others have adopted a forced approach, prioritizing a specific outcome (emancipation) over allowing it to potentially emerge organically from students' own initiatives and questions – thereby risking the compromise of student autonomy and student direction. This could explain why the project work, as envisioned by Hultengren (1979, 2014), did not succeed.

3.2.1 The two aspects of education and a Danish interpretation of *Bildung*

For interdisciplinarity in project work to emerge organically from students' *interests* and their identification of the problem and the research question, we must develop a more nuanced understanding of the roles of both teachers and students in IPP. Though we agree with the starting point and movement of Negt's (1971) principle of exemplarity from the particular to some broader understanding or context, we need to introduce pedagogical distinctions between *instruction* and *Bildung* (Kemp), and between *teacher directed* and *student directed* activities (Rousseau), in order to identify the didactical side of the principle, and the *Bildung* side, and the different ideals that can be connected to the educational activity by teacher *and* student. Building on the discussion of *Bildung* below, we return in Section 3.2.3 to offer a reinterpretation of the principle of exemplarity.

Unlike Hultengren (2014), Negt (1971, 1994) and Illeris (1974, 1991), we argue that student-directed activities in higher education (HE) within the human and social sciences should be distinguished from teacher-directed activities to emphasize the fundamental differences in role and responsibility. As we will show, this distinction has consequences for both the instructional aspect and the *Bildung* aspect of education in HE.

Who determines the *Bildung* ideals within education? While Negt also distinguishes between the development of competencies and *Bildung,* and emphasizes the importance of "the development of an individual world view" (Negt, 1994 cf. Nielsen, 1997, p. 271), he does not empower students to identify their own *Bildung* ideals.

Following the Danish philosopher of education Peter Kemp (2011, p. 27), education encompasses both *instruction* (in German, *Heranbildung*, in Danish, *uddannelse*), which is the technical side of education emphasizing competencies, and *cultivation* (in German, *Bildung*, in Danish, *dannelse*), which is the ethical-political side of education emphasizing independent thinking. In his educational-philosophical analysis of the meaning of the concept of education (from Latin *educare,* meaning "to take care of"), education today refers to "two different aspects […]: cultivation (*Bildung*) and instruction. Cultivation makes people cultural beings, while instruction makes them knowing beings. Cultivation has an ethical goal, while instruction aims at competence" (Kemp, 2011, p. 151). Education should both instruct *and* provide rich and free conditions for human development. There is no conflict between the two: *Bildung* is dependent upon competencies and deep disciplinary knowledge, and vice versa, according to Kemp (2011, p. 152).

To understand the unique Danish interpretation of *Bildung,* we must first trace the historical development of the concept. Gustavsson argues that, until around 1800, *Bildung* "was informed by two key elements – *a free, endless process which originated from the Greeks, and a picture of the ideal, Imago Dei borrowed from Christian mysticism*" (2014, p. 111). *Bildung* referred to the formation and the self-formation of the individual in the light of the shared tradition and culture (Kemp, 2011, p. 185). Within the Danish educational tradition of the eighteenth century, Kemp contends that *Bildung* came to be seen as an integral part of education at all levels - from primary school, technical school, university and folk high school. It was intrinsically linked to the entirety of an individual's life and not confined to the elite (2015, p. 19). *Bildung* in the Danish tradition was thus understood as a free process, open to a multitude of ideals or "results" within the commonly recognized culture. In summary, a Danish interpretation of *Bildung* can be described as the cultivation of the individual in the light of an ideal from the commonly recognized culture, which contains many different ideals.

This distinction (instruction vs. *Bildung*), we argue, grants students a double freedom (cf. section a and b below) and assigns a more humble role to the teacher, who must now acknowledge power dynamics and the importance of stepping aside as a teacher (cf. Kemp, 2011; Rousseau, 1762/1979).

3.2.1.1 Double freedom a) The freedom to choose *Bildung* ideals

Bildung ideals are not predetermined by the teacher, the philosopher, the tradition, or any other collective institution in society; rather, it is no longer – as in the original meaning of *Bildung* – defined by default as "*das Bild der Gottheit*" (the image of divinity, Herder), freedom (Rousseau), autonomy (Kant), "elevation" to *Allgemeinheit* (Gadamer), social responsibility (Kemp), or emancipation and imagination of a

better world (Negt). *Bildung* in the Danish tradition refers to free processes of understanding and identifying ideals and values in the hands of each student *as an individual*, exposed - of course - to a wide spectrum of cultural role models (e.g., literature, teachers), pedagogical ideals, and historical traditions within culture and history. According to Humboldt, for *Bildung* to take place, the individual must have freedom and independence from the state (Kemp, 2015, p. 23). We argue that this must also include, freedom and independence *from the teacher and the educational institution* (cf. the symmetrical roles of Table 2).

Kemp (2011) analyses the role of the teacher and the student. According to Kemp, cultivation and instruction place a great responsibility on the teacher, but the teacher must also aim to

TABLE 2 Roles, responsibilities, power and (inter)disciplinary knowledge in IPP.

Asymmetrical roles: instruction		Symmetrical roles: Bildung
Student and project group	*Supervisor*	*Student & Supervisor*
A) The project group works student-directed and problem-oriented and thus has the *power to choose and is responsible for all choices* in their project work. This includes the problem-identification, research question, research design, philosophy of science, theories and methods etc. in the report, and that these choices are within the thematic framework of the project module (cf. pedagogical and epistemological aspects).	B) The supervisor steps aside from the directing role but actively listens to students' own decision-making and supports them through challenging their thinking and sharing ideas, scientific literature, and resources that can help the group realise their research question within the thematic framework. The supervisor reads and comments on drafts sent by the project group (cf. pedagogical and epistemological aspects).	Although there is an asymmetry in the power relations between students and supervisor(s) (representing the HE institution) within the instructional aspect, they play symmetrical roles when it comes to the *Bildung* aspect of education. We distinguish between personal-professional, ethical-political, and pedagogical *Bildung* ideals. We associate IPP with the pedagogical ideals of both emancipation *and* envisioning a better world (utopian futures), beginning with the way the project group and supervisors respect the symmetrical and asymmetrical division of roles, powers, and responsibilities. Although students and supervisors must strive for a common understanding of IPP didactics within the instructional aspect of education, they must also cultivate a pluralistic *Bildung* culture within the institution through respect for the diverse ideals and values of each individual. Pedagogical ideals should be openly debated within the educational institution and among educators; yet these cannot be imposed upon students as part of the teacher's or institution's instructional authority. *Bildung* processes and ideals must be freely embraced by the individual student and supervisor (cf. the dual movements of the *Bildung* process). A reversibility exists between students and teachers, in that teachers can also adopt ideals or styles of thinking from students through creative mimesis.
B) Students actively listen to the supervisor's feedback and ask clarifying questions in light of their pre-understanding, guiding the supervisor to address their insufficient understanding specifically and questions with respect to their research question and inquiry in general (cf. epistemological aspect).	A) *The supervisor acts as the disciplinarian* and the guardian of disciplinary norms and instructs the students on research design, philosophy of science, theory and method in light of their research question. However, the supervisor advises on academic questions that fall outside students' academic progression.	
A) As part of the brainstorming, group formation and preparatory phase of the inquiry, students develop their problem identification, research question and inquiry within a free space through their interests (pedagogical aspect) and creative play and thinking with each other and the supervisor within a potential space (interactional aspect). The resulting research question *often calls for sociological imagination and other interdisciplinary approaches by itself* (epistemological aspect).	B) The supervisor actively proposes (inter)disciplinary perspectives and develops reflective questions for the group, based on the IPP principle of exemplarity and possible knowledge interests within the thematic framework (cf. epistemological aspect). The supervisor creates a free space for wondering as well as a potential space, engaging with the students' potentials and ideas "as if" they are researchers and anything is possible (pedagogical and interactional aspects).	
B) Students prepare the agenda and drafts for supervision and search for relevant scientific publications. Students compile a reading list as part of the project report. However, students are dependent on the supervisor, who must approve the list based on the thematic framework of the module. The supervisor also serves as the oral examiner during the group examination, with an internal or external examiner present (pedagogical aspect).	A) The supervisor serves as *the pedagogical authority for organizing learning as group-organized, problem-oriented project work*, responsible for teaching during supervision and holding the oral group examination (cf. the learning outcomes of the project module), upholding student responsibilities and freedoms in accordance with the didactics of IPP (including student-direction and exemplarity) (pedagogical aspect).	
A) Students are responsible for all processes related to upholding the group as a working group (Bion) and producing the final report, including balancing expectations, process dialogues, conflict resolution and general process orientation (interactional aspect).	B) The supervisor strives to be sensitive to each student's lifeworld and the group's dynamics and the internal collaboration, to understand what holds them back from functioning as a work group and deepening their individual understanding (interactional aspect).	

Regarding the asymmetrical roles, A) signifies a foreground role of power and responsibilities, while B) signifies a background role, yet still plays an active role in relation to the adjacent horizontal rows (symmetrical and asymmetrical columns).

enable students to take responsibility for their own lives and ideals, introducing a dimension of mutuality or reversibility between student and teacher roles. Kemp uniquely describes this as mimesis, a concept we revisit later (cf. Section 3.3.1) (Kemp, 2015, p. 158). For now, we follow Kemp's analysis of the student-teacher relation regarding mimesis in the *Bildung* aspect of education: "*we choose our own master or masters freely but not, as it is often assumed today, on our own terms*" (Kemp, 2011, p. 196, italics in original). Kemp, in other words, refutes the notion that anybody cultivates themselves on their own terms; *Bildung* is social: it emerges from the commonly recognized culture, something "other" (alien) to the individual (2011, pp. 185–188; 2015, p. 47). No one can cultivate themselves in their own image.

To emphasize the existential and individual selection of one's own variation of ideals and one's own voice, Feilberg (2014) suggests understanding the *Bildung* process in two movements, inspired by Gadamer's "basic idea" of *Bildung*[3] and Kemp's interpretation of mimesis as creative imitation: "Initially, I should try to be like the master and, then, I should try to do it a little better" (2011, p. 196), whereby "[w]e educate ourselves through others" (Kemp, 2011, p. 245).

The first movement entails the self-forgetting absorption and involvement with something "other" (alien) that speaks to us, whereby– in an existential sense – we adopt aspects of that ideal model or style of thinking (e.g., a tradition, *Bild*). But this alone is not *Bildung*. *The second movement* involves the creation of a critical distance and the taking oneself back to develop a voice of one's own by forming an independent judgment about what was pre-reflectively adopted (Feilberg, 2014, p. 96, 102). The first movement remains unattainable if one acts only strategically within the educational arena, is unable to immerse oneself in something for its own sake, or dares not engage and thus put one's *existence at stake* (Feilberg, 2014, p. 111; 2019, p. 135). The second movement of the *Bildung* process is unattainable if one cannot critically assess, establish distance, and form independent judgments when reflecting on what was uncritically adopted in the first movement. We argue that both the student and the supervisor are free to live by their identified ideals and values, but these cannot be imposed on others. Each embodies and cultivates the ideals and values important to them in their context, in light of the broader culture. This understanding represents a break with the classic German *Bildung* theories of Gadamer and Negt (Kemp, 2011, p. 206).

3.2.1.2 Double freedom b) A wide thematic framework encourages both *Bildung* processes and interdisciplinarity during project work

According to Gadamer, "*Bildung* as such cannot be a goal; it cannot as such be sought, except in the reflective thematic of the educator" and *Bildung* "has no goals outside itself" (1960/2006, p. 10). We argue, however, that the formal instruction aspect of education can be organized in ways that informally invite *Bildung* processes within the curriculum, in case students choose to pursue them. Likewise, interdisciplinary inquiries are best encouraged through project modules with a wide thematic framework – that is, modules that do not explicitly call for interdisciplinary inquiries in, e.g., their learning outcomes. We argue that – paradoxically – presenting students with wide thematic frameworks gives *Bildung* processes and interdisciplinary inquiries the best conditions to emerge in authentic ways, even though they might also not emerge at all. Here is why: starting with *Bildung* processes and then interdisciplinarity in project work.

As Rousseau (1762/1979, p. 119) argues, there is no way a teacher can force cultivation and independence onto a child or a student: teaching is "a difficult art" that they must perform "without precepts". Rousseau identifies the opposite of independent education of a pupil as follows: "Submitted in everything to an authority which is always teaching, yours does nothing unless given the word. […] Soon he will dare to breathe only according to your rules. About what do you want him to think when you think about everything for him?" (Rousseau, 1762/1979, p. 118). Instead, Rousseau advises: "Leave him alone at liberty. Watch him act without saying anything to him" (1762/1979, p. 161). As the first movement of the *Bildung* process according to IPP requires existential involvement by the student, Rousseau's analysis of the role of freedom in education is relevant. We find inspiration in Rousseau's discussion of how the teacher must grant the "suitable" amount of freedom in accordance with each "age, each condition of life" and the "maturity" of each individual student (Rousseau, 1762/1979, p. 158; see also Kemp, 2011, p. 170).

Furthermore, Rousseau points to a double approach to education. He points to both the importance of cultivating the wild independence of the child from childhood, and the importance of formal instruction in school and beyond. This is captured in Rousseau's paradoxical argument: "You will never get to the point of producing wise men if you do not in the first place produce rascals. This was the education of the Spartans: instead of being glued to books, they began by being taught how to steal their dinner" (Rousseau, 1762/1979, p. 119). In a wider sense than Negt's concept of free spaces within education (*frirum*), we argue that Rousseau shows that by learning from a young age about the free space within every field, the child can learn to *exercise their freedom* (not to steal). In Section 3.3.3 below, we will, with Winnicott, introduce the concept of the potential space, which the adult creates with the child for play and developing the child's independent thinking. With respect to encouraging *Bildung* processes in a HE context, informally, we find grounds in Rousseau and argue for the importance of creating free and potential spaces for students alongside the formal curriculum, so that they can exercise their freedom to choose, become involved and interested, and thus put their *existence at stake*, - or choose not to - as students can only learn to exercise their freedom and pursue *Bildung*, if they are free to dismiss it, as it has no goals outside itself (cf. Gadamer). Based on Rousseau's work, Thøgersen (2010, p. 51) distinguishes between self-direction (in Danish, *egenstyring*) and foreign direction (in Danish, *fremmedstyring*). Pedagogy, as defined by Thøgersen (2010), is the formation of the human being according to specific value orientations through education. Pedagogical

[3]Based on Hegel, Gadamer formulates the "basic idea" of *Bildung* as follows: "To recognize one's own in the alien, to become at home in it, is the basic movement of spirit, whose being consists only in returning to itself from what is other" (1960/2006, p. 13).

theories such as IPP are normative by nature and rely on certain assumptions about the human being, which must be critically and openly reflected upon (Thøgersen, 2010, p. 52). Within IPP, *Bildung* is understood as something that can only come about in freedom and emerge in free spaces, and within the educational field, each person (student or teacher) must regard the other with respect for the self-direction of the other. If a supervisor in IPP is invited by the student into a dialogue about the student's *Bildung* processes, it is understood to be a dialogue from a symmetrical position (cf. Table 2).

Concerning the encouragement of interdisciplinarity in project work as part of the instructional aspect of education, the work of Rousseau offers further pedagogical distinctions to IPP. Continuing the analysis of the role of freedom in education from above, Rousseau argues that when raising a child, the most important instrument is "well-regulated freedom", which allows the adult to guide the child "where one wants by the laws of the possible and the impossible alone" (Rousseau, 1762/1979, p. 92). Furthermore, we argue that AAU has developed a culture and a pedagogical understanding of tailoring the wide thematic frameworks in interdisciplinary project modules, which open up countless problems and research questions for students to identify and explore, and are thus pedagogically and didactically designed with a "suitable" amount of "well-regulated freedom" (Rousseau) with respect to student's disciplinary and interdisciplinary "maturity" and progression within the study programme. Within IPP, interdisciplinary project work can only succeed with a combination of freedom and support. In his works, Rousseau describes how self-formation is made possible through well-regulated and tailored freedom, in combination with a teacher who acknowledges the power dynamics within teaching and reflects pedagogically on her role when supporting self-directed activities or employing foreign directed activities (Thøgersen, 2010). The same is true of IPP. When encouraging interdisciplinarity in project work within IPP, it must be based on a culture that provides both freedom *and* support in order to *invite* students to formulate their own research questions and design their own inquiries from the very first semester onward.

We argue that well-regulated freedom is the most powerful way to invite both *Bildung* processes and interdisciplinarity into problem-oriented project work. The truly challenging part is how to support these processes within IPP, which is the theme of the following sections.

3.2.2 The interdisciplinary potential of the wide thematic framework in IPP

Project work at AAU started in 1974 with project catalogues and teacher-identified problem areas. Though it was possible for students to develop and have their own subject approved by the study board, most student projects were highly framed. Hultengren (1979) for instance describes a project group who had been assigned the local shipyard and its management team as a point of reference for their problem-oriented project work.

But over time supervisors developed a better way of offering freedom to students through the didactic concept of *a wide thematic framework* (*temaramme*). Within IPP, the thematic framework defines the overall problem area and the permitted disciplinary fields and knowledge areas of the project module. Consequently, students exercise their freedom to choose a topic and develop a research question for their project within the thematic framework described in the curriculum of the module. Cf. Section 3.2 and the exemplary breadth-and-depth model of IPP, students are introduced to key literature and discussions within the disciplinary field as well as dominant research methods before selecting their own project topic. The implications for interdisciplinary work are that the more deeply students are involved in the process of problem orientation, problem identification, and problem analysis, in light of their own interests and horizons of experience (lifeworld), the more complex the problems they will undertake and the greater their motivation for - and need for - interdisciplinary approaches to inquiry. As doing interdisciplinary research is twice the workload, the motivation for pursuing this work is vital.

Finally, as a consequence of student direction in IPP, problem-oriented project work is not expected to follow an emancipatory knowledge interest, as Illeris (1974, 1981, 1991) and Hultengren (1979, 2014) originally prescribed. Back then, students were expected to unfold broader societal relations in all project reports within the human and social science studies at AAU, according to Hultengren (2014). Instead, within IPP, we follow Habermas' epistemological distinction between three knowledge interests: the practical-understanding, the technical and the emancipatory knowledge interest (Habermas, 1978). According to this epistemological perspective of IPP, students are given the *opportunity* to pursue an emancipatory knowledge interest in their project work should they choose so, and this can be seen as a unique *potential* of problem-oriented project work; however, depending on the thematic framework of the project module, a practical-understanding and a technical knowledge interest are just as viable within problem-oriented project work (Feilberg, 2022b, pp. 85–86). The philosophy of science and the introduction to several research paradigms, with ontological and epistemological assumptions and methodologies explicitly stated, are also a key quality of IPP, as they offer the necessary epistemological theories for conducting interdisciplinary inquiries in a scientifically reflective way.

3.2.3 A reinterpretation of the principle of exemplarity for IPP

The principle of exemplarity is important when inviting greater levels of student freedom into an educational activity such as interdisciplinary project work. Based on the analysis and distinctions presented above, we suggest the following reinterpretation of Negt's (1971) principle of exemplarity within IPP.

Concerning the instruction aspect of education regulated by curriculum and learning outcomes: The didactical working process of IPP is that of working from the particular to broader, exemplary relations within curriculum, when students work from a problem area and over time identify a specific problem and research question and build an inquiry from there. These *pedagogical aspects* of project work also include student direction, problem orientation, project organization and interdisciplinarity, which must be shared concepts for students and supervisors alike. The *epistemological aspect* of project work

within IPP concerns the epistemological process of scientifically inquiring into the research question by a selection of disciplinary and interdisciplinary theory and methods. The *interactional aspect* of project work within IPP concerns the collaboration within the project group and with the supervisor and external partners. Both the epistemological and the interactional aspects are described in the curriculum of the project module as learning outcomes in broad terms, together with a short and wide description of the thematic framework of the module.

Concerning the *Bildung* aspect of education: Any educator is free to choose their own ideals, but we suggest that IPP is based on the following pedagogical values. The principle of exemplarity consists of three dimensions and they relate to the *Bildung* aspect of education, and thus serve as invitations to the students, and as dimensions that the educator can strive to strengthen when students choose the research question, the problem and design their inquiry: 1) the proximity to the student's individual interests and experienced conflicts[4] 2) the potential to explore broader societal and (inter)disciplinary relations, 3) the potential to challenge students to cultivate critical thinking (emancipation cf. Negt, 1971) and to imagine a better world (utopian futures, cf. Negt, 1994; Nielsen, 1997).

Of course, the *Bildung* ideals that we have proposed here can be changed by any other supervisor with respect to their own practice – *Bildung* ideals are up to each educator to reflect on and decide. Educational institutions can try to foster a certain culture or introduce certain learning methodologies, but then they have the same responsibility as that of a teacher to a student: the institution can invite certain ideals and values (*Bildung*), but the teacher and the student are free to make up their own mind.

3.3 Qualities of the interdisciplinary, problem-oriented project pedagogy

We summarize the remaining qualities of IPP and present an overview in Table 2, which describes the principal roles and the reversibility of roles and responsibilities.

Interdisciplinary work progresses from narrow interdisciplinarity in project work within the programme, which later lays the foundation for broad interdisciplinary inquiries and research questions in e.g., the master's thesis. Below, we present pedagogical distinctions and terminology which are helpful when supporting interdisciplinary project work within IPP.

3.3.1 Mimesis and reversibility of the student and supervisor roles

Mimesis within education describes how imitation is a powerful natural part of learning: mimesis according to Merleau-Ponty "is the power of assuming conducts" of others, both pre-reflectively and consciously, of style of thinking and practical skills (1964 cf. Feilberg, 2014, p. 105). Mimesis is part of both instruction and *Bildung* (Kemp, 2011, p. 196). Even though we certainly adopt styles of thinking and doing from others to a great extent[5], mimesis in a *Bildung* context also entails acquiring a form of independence in relation to them (Feilberg, 2014, p. 107).

Ricoeur recounts how he learned philosophy: "I read Sartre, Merleau-Ponty and others, *and I wanted to do it a little better*" (Kemp, 2011, p. 187). Critique of that which is imitated (mimesis) is a necessity for *Bildung* processes to unfold, and self-critique is at the heart of any master, argues Kemp (2011). A person imitating others without recognizing the source is a clone. To emphasize this difference, Kemp speaks with Ricoeur about *creative mimesis*: "creative mimesis breaks through the self-enclosure of the subject without enlisting it in the anonymous masses of rank and file. […] The teacher forms the student, but the student forms him or herself through the choice of the teacher" (2011, pp. 196–197). There is an element of seduction through what Kemp describes as true authority, which can initially captivate the student.

The theory of mimesis (cf. Kemp, 2011) can also describe the reversibility of the supervisor and student roles. What is imitated can be a person, a style of thinking etc., though Kemp uses the teacher and student relation as an example. According to Kemp, "The best teacher is the one who can make his or her student autonomous" (2011, p. 197). Kemp describes an ideal of a mimesis relation between teacher and student, which is possible only if they uphold a distance and a mature way of interacting that holds jealousy (and we add, narcissism) at bay. This ideal mimetic relation, Kemp continues, "is only possible if I, as the teacher, do not want my student to repeat everything I do and, to the contrary, hope for and invite a constructive and challenging student-teacher relation, so that the student can become my teacher, and if I, as a student, do not regard the teacher as infallible, believing that I cannot do better than him or her and make even small improvements" (Kemp, 2011, p. 194). We argue that supervision within IPP can strive for such reversibility in roles between student and supervisor, but at the same time we identify principal roles and responsibilities for the student and the supervisor (cf. Table 2).

3.3.2 Roles and responsibilities as student and supervisor

Following Visholm (2004, pp. 43–45), we ascribe different formal roles[6] to the individual student and the supervisor according to their task, authority, and responsibilities. The primary task of the students who form the project group (the work group) is to organize and focus the work on the primary task: identifying and inquiring into the problem and the formulated research question within the wide thematic

[4]Contrary to Hultengren (1979, 2014), we argue that students do not need to have experience from their own life for it to count as a "conflict" in dimension 1. An example is foreign wars and disasters of which students can identify as part of their wider lifeworld (Feilberg, 2014, 2022a, p. 45).

[5]Crediting the ideas of others is an academic necessity, however, this does not change the fact that the most powerful way of learning is through the imitation of others (Dreyfus, 2001, p. 48).

[6]In Illeris' work, students and supervisors "have different functions and responsibilities" (1991, 46), although we refer to them as roles.

framework of the project module. We must distinguish between person and role, and between the formal role and the dynamic processes that are active on an unconscious level in the individual, the project group and the supervision group (student and supervisor) (Feilberg, 2015). Within the project group (students working on the project independently), students draw on knowledge presented in courses but also on the collective pool of knowledge resources that group members bring to the table qua their diverse lifeworlds, including their social and cultural backgrounds, for example mastery of several languages, involvement in civil society organizations, travel experience, cultural understanding, useful contacts and networks, consumption of a wide variety of media, specific interests etc. As identifying the research problem implies activating diverse academic and non-academic knowledge resources, processes of peer learning form an integral part of IPP (Jæger, 2020).

Within the supervision group, we distinguish between the roles of students and supervisors. Table 2 above lists asymmetrical and symmetrical roles. While the students have the freedom and power to decide on the epistemological aspects of their project work within the thematic framework, as part of the pedagogical principles of student direction and problem-orientation, the supervisor must actively facilitate this by pedagogically holding students to their responsibilities, while also taking seriously one's own role as epistemological teacher, guardian of disciplinary norms (cf. powerful knowledge), and interactional supervisor. As a supervisor, one must investigate what holds students back from a deeper understanding of theory and method (epistemological aspect) and improved collaboration (interactional aspect), while addressing the invisible and problematic "professorial power" that some students perceive as lying in the hands of the supervisor as an "authority on truth and the best research topics" (cf. Feilberg, 2015; Hultengren, 1979). The supervisor's most important skill involves identifying when to intervene in collaboration problems, when to instruct in disciplinary knowledge, and when to let the process remain entirely in the hands of the project group.

3.3.3 The potential space and holding as key qualities of IPP

The supervisor plays an important role in creating a supportive space and identifying what each student dreads and avoids as well as what holds them back from their independent development and progression in light of the curriculum and *Bildung* processes. The psychology of supporting the child on its journey into the world has been examined by the psychoanalyst Winnicott, and his work is also relevant to IPP. He introduces the concept of potential space in *Playing and Reality* (1971) as part of understanding the process by which a child develops the capacity to be alone and to develop his or her own personal space for thinking and acting independently. As part of this process, the child and adult momentarily create an undifferentiated experience, intermediate between fantasy and reality, which supports the unfolding of creativity, dreams and play, and where the child gradually discovers himself or herself as both separated from and connected to others (ibid.). This process in the potential space takes place between a child and an empathic "other", who is sensitive to the state of mind of the child. In this interaction, the intermediate area can unfold, where the child safely investigates new potentialities for creative thinking and discoveries (Ogden, 1990, p. 182, 203). The transitional object (e.g., the favourite teddy bear) is what supports a child in taking a journey otherwise impossible.

We propose that creative and independent interdisciplinary student projects involve a project supervision characterized by a Winnicottian function of holding (or container–contained process) and a willingness to join the student in a potential space—an intermediate area of experience—where the supervisor imaginatively enters the student's experiential world. For example, the difficulties they may experience include hesitation in choosing the research question independently and engaging with and applying complex theories, methods, and philosophy of science. In this potential space, especially in the brainstorming and preparatory phase, new interdisciplinary ideas and approaches are tried out in a playful manner, and students can try out ("as if") potentials, styles of thinking and different roles, such as: the role of the researcher imagining new questions, new interdisciplinary inquiries, or styles of thinking such as "thinking philosophically about an issue like Merleau-Ponty". As we saw in the theory of mimesis, the power of assuming conducts or styles of thinking can be imitated from teacher to student. Over time, the students will embody the creative play with the new roles and styles of thinking by integrating them into their own habitus (Feilberg, 2019). This potential space, we argue, is especially important to interdisciplinary inquiries, where new questions are raised and new ways of inquiring into them must be developed creatively. Following the potential and shared space, separateness emerges as students and the project group after supervision are on their own and try to materialise the project; this brings frustration, and in time, an independent and cultured habitus. It is important to help contain this frustration as a supervisor. In the end, the project report can act as the transitional object, which in a psychological and pedagogical sense supports students in transitioning from novices to defending their work and acting as "experts" within the subject and their exemplary project, fully on their own during the oral examination.

4 Discussion

The evolution of IPP at AAU illustrates how an institution, its departments, and its programs move between discipline-oriented and interdisciplinary logics as they adapt to external and internal pressures and renew themselves to respond to social, economic and political circumstances. On the one hand, the IPP model granting free space for interdisciplinary student research faces constant pressures as the university needs to fulfill national expectations regarding academic content, student-teacher contact (calculated as hours spent with a teacher and therefore most effectively achieved through lecture-based teaching), and economic efficiency. Cuts in supervision resources make programs encourage students to align their choice of project topics and research questions more directly with program-defined outcomes and supervisor expertise (e.g., by project catalogues), often leading to higher discipline orientation.

Isolated events such as the rebranding of the former Politics and Administration program to Political Science and the ensuing downscaling of project work at the request of other Danish universities illustrate how the necessity of adapting to existing institutional structures at national level results in a reduction of space for students' interdisciplinary explorations (Klindt, 2025). Such developments illustrate that higher discipline-orientation does not necessarily happen because decision-makers view disciplinary knowledge as inherently "better" as the powerful knowledge approach would argue. Instead, steps towards higher discipline-orientation comply with intentions to achieve compatibility with national and international curricular standards (such as the three-cycle system) and to strengthen the competitiveness of the university's programs.

On the other hand, interdisciplinarity is flagged as a significant strategic priority (AAU, 2021). The university has recently supported interdisciplinary program collaboration, for example in its SSH-STEM integration project (IAS-PBL, 2026). Interdisciplinarity is highly valued but in addition to a strong base of disciplinary knowledge. Consequently, study activities designed to foster interdisciplinary (cross-program) collaboration ["loud interdisciplinarity" (Lindvig, 2022)] tend to appear in later semesters and require students to actively choose the interdisciplinary option. Interdisciplinary learning experiences may, as a consequence, only appeal to some, and most likely the strongest students.

Facing an emergent bifurcation between increasing discipline-orientation and "staged" or "loud" interdisciplinary learning activities, IPP offers a "third way", a way of making interdisciplinary experience a part of every student's educational path from the first semester. Above, we suggest how interdisciplinarity can be promoted within programs that are oriented towards specific disciplines or established interdisciplinary fields featuring IPP. Sociological imagination and the creation of free spaces (cf. Negt) contribute to the epistemological aspect of IPP, while the potential space and holding (cf. Winnicott) contribute to the interactional aspect of realizing IPP. Finally, reflecting on power dynamics within education (cf. Rousseau), as well as the normative ideals of pedagogical practice, contributes to realizing the pedagogical aspect of IPP.

In our analysis, key qualities supporting interdisciplinarity in education include: A) granting individual academic freedom to each student within curricula, e.g., in the form of a wide thematic framework during student-directed activities. At the same time, the supervisor must insist that students exercise their well-regulated freedom (cf. Rousseau) and conduct their own problem identification and research question (pedagogical and epistemological aspect of IPP). The supervisors must also take care of (*educare*) and nurture a *Bildung* culture and the *Bildung* aspect of education – for example, by setting the individual free to identify their own *Bildung* ideals and masters through creative mimesis (cf. Kemp) and the dual movements of *Bildung* (cf. Feilberg), as this invites students to pursue meaningful interdisciplinary, and independent academic work. When an individual is recognized by the "other" as free, responsible and as having embarked on their own *Bildung* journey, it can open up opportunities to students *and* teachers to engage in education – not for grades, fame or personal income, but – as an end in itself (cf. Gadamer), that is existentially meaningful and socially responsible in light of the shared lifeworld and the social community (cf. Kemp). B) developing an interactional culture among supervisors, which supports students during supervision and invites them into a potential space of creative play with ideas, potentials and reversibility with the advanced roles and responsibilities that make interdisciplinary inquiries possible (interactional aspect of IPP). The role of the supervisor is a difficult art of stepping aside and supporting transitions, mimesis and reversibility between symmetrical and asymmetrical roles along the process of supervision (cf. Table 2). This includes supporting the creation of a "free space" for exemplary and interdisciplinary problem orientation and a "potential space" of creative playing and creation within the project group. As shown, Winnicott's concept of holding – by creating an emotionally supportive atmosphere as supervisor – is essential for students' creative exploration of new ground within their project work – both with respect to competencies and knowledge (instruction) and to finding their own voice (*Bildung*).

5 Concluding remarks

This article has explored two different theoretical perspectives for thinking about disciplinarity and interdisciplinarity in the curriculum – the "powerful knowledge" approach, emphasizing the value of disciplinary knowledge and the German and Danish "*Bildung*" tradition, which requires combinations of deep "alienating" engagement with disciplines as well as experiences of interdisciplinary openness. We particularly analyzed the development of problem- and project-based learning at AAU with a view to how this form of teaching and learning has oriented itself vis-à-vis these two theories. During the first years of the university's existence, powerful knowledge thinking – according to our analysis - did not hold much sway even if it erupted occasionally in teachers' concerns regarding students' disciplinary knowledge. Instead, interdisciplinary openness, social relevance, and focus on student interests and direction were institutionalized in the university's interdisciplinary "basic years". External and internal pressures have contributed to narrow spaces for interdisciplinarity in projects, particularly as the "basic year" disappeared and the time formerly spent on interdisciplinary student research on broadly defined thematic frameworks was reclaimed by the university's programs. To build on the insights of 50 years of experimentation with interdisciplinarity at AAU, we identify the key qualities of Interdisciplinary Problem-oriented Project pedagogy (IPP).

A significant insight gained in this article is that whereas powerful knowledge thinking gains traction institutionally, students' right to determine the content and direction of their projects persists (as per the new AAU, 2025 PBL principles). At the level of everyday practices and interactions, this requires supervisors, on the one hand, to facilitate students' free and group-directed epistemic travels into and across disciplines, and on the other to be "guardians of disciplinary knowledge". To enact this double role at the interface of student interests and disciplinary standards and norms takes a much more precise and detailed understanding of supervisor-student interactions and a sensibility towards power structures in these interactions, which we have illustrated above. Ideally the supervisor empathetically follows and nurtures students' interests, which often defy

disciplinary boundaries, and support them in their idea realizations that align with disciplinary, program, and institutional norms. The current form of supervised, student directed project work largely fulfills the criteria for interdisciplinary problem-oriented project work established above (cf. exemplarity, student direction, supervisor aware of power dynamics, broad thematic framework; supportive institutional framework). However, the contradictory pressures placed on supervisors in the realization of interdisciplinary openness and a stronger emphasis on mastery of disciplinary knowledge are probably not fully realized.

Problem- and project based learning at AAU is founded on the principles of problem orientation and exemplarity. Although the critical theory version of these concepts never took root at AAU to the extent that it did at its sister university in Roskilde, the concepts, in re-negotiated forms still influence the development of project and problem-based learning at AAU. At the heart of these concepts lies an interest in developing an ability to connect knowledge and reality, thinking and experience and in supporting *Bildung* and social responsibility. Educating with a point of departure in the lifeworld experience of students while enabling deep theory-based understandings of social, cultural political and technological problems requires curricula that combine disciplinary knowledge with spaces for interdisciplinary explorations.

Author contributions

CF: Conceptualization, Writing – original draft, Writing – review & editing. KJ: Conceptualization, Writing – original draft, Writing – review & editing.

Funding

The author(s) declared that financial support was not received for this work and/or its publication.

Conflict of interest

The author(s) declared that this work was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

Generative AI statement

The author(s) declared that generative AI was not used in the creation of this manuscript.

Any alternative text (alt text) provided alongside figures in this article has been generated by Frontiers with the support of artificial intelligence and reasonable efforts have been made to ensure accuracy, including review by the authors wherever possible. If you identify any issues, please contact us.

Publisher's note

All claims expressed in this article are solely those of the authors and do not necessarily represent those of their affiliated organizations, or those of the publisher, the editors and the reviewers. Any product that may be evaluated in this article, or claim that may be made by its manufacturer, is not guaranteed or endorsed by the publisher.

References

AAU. (2021). *Knowledge for the world 2022-2026*. Aalborg: Aalborg University. Available online at: https://prod-aaudxp-cms-001-app.azurewebsites.net/media/odnnfqrx/aau-strategy-2022-26.pdf (Accessed August 5, 2026).

AAU. (2025). *Aalborg University Principles for Problem- and Project-Based Learning*. Aalborg: Aalborg University. Available online at: http://www.en.aau.dk/about-aau/profile/pbl (Accessed August 5, 2026).

Bernstein, B. (1999). Vertical and horizontal discourse: an essay. *Br. J. Sociol. Educ.* 20 (2), 157–173. doi: 10.1080/01425699995380

Christensen, G. (2020). Grupper i pædagogikken: fra indbyrdes undervisning til progressive tiltag og cooperative learning. [Groups in pedagogy: from mutual teaching to progressive initiatives and cooperative learning]. *Nord. Tidsskr. Pedagog. Krit.* 6, 1–15. doi: 10.23865/ntpk.v6.1635

Coghlan, D., and Holian, R. (2021). Insider action research as leadership-as-practice: a methodological reflection for OD scholar-practitioners. *Organ. Dev. Rev.* 53 (5), 13–17. Available online at: https://hdl.handle.net/10779/rmit.27553512 (Accessed April 19, 2026).

Dirckinck-Holmfeld, L. (2002). "Designing virtual learning environments based on problem oriented project pedagogy," in *Learning in Virtual Environments*, eds. L. Dirckinck-Holmfeld, and B. Fibiger (Frederiksberg: Samfundslitteratur), 55–79. Available online at: https://vbn.aau.dk/files/648315714/POPP020102fin.pdf (Accessed August 5, 2026).

Dreyfus, H. (2001). *On the Internet*. New York: Routledge.

Feilberg, C. (2014). *Dannelsen af en Psykologisk og Videnskabelig Habitus hos Psykologistuderende. [The Formation of a Psychological and Scientific Habitus in Psychology Students]*. Roskilde: Academic Books.

Feilberg, C. (2015). Selvrefleksion som uddannelsesgreb – en kritisk diskussion. [Self-reflection as an educational grip – a critical discussion]. *Nordiske Udkast* 2, 42–61. doi: 10.7146/nu.v43i2.141468

Feilberg, C. (2019). "In deep water: university students' challenges in the processes of self-formation, survival or flight," in *Educational Dilemmas: A Cultural Psychological Perspective*, ed. L. Tateo (London: Routledge), 126–138. doi: 10.4324/9781315101095-8

Feilberg, C. (2022a). "Problemorienteret projektarbejde og politisk dannelse. [Problem-oriented project work and political *Bildung*]," in *Projektarbejdets Dannelsespotentiale. [The Bildung Potential of Project Work]*, Studier i problembaseret læring ved videregående uddannelser eds. C. Feilberg, I. G. Bo, and A. Q. Romani (Aalborg: Aalborg Universitetsforlag), 25–56.

Feilberg, C. (2022b). "Problemorienteret projektarbejde, PBL og Aalborg-modellen. [Problem-oriented project work, PBL and the Aalborg model]," in *Projektarbejdets Dannelsespotentiale. [The Bildung Potential of Project Work]*, Studier i problembaseret læring ved videregående uddannelser. eds. C. Feilberg, I. G. Bo, and A. Q. Romani (Aalborg: Aalborg Universitetsforlag), 59–96.

Feldt, J. E. (2023). Exemplarity as deliberative curriculum: finding out what to study, why, and how. *Stud. High. Educ.* 48 (3), 399–412. doi: 10.1080/03075079.2022.2138850

Flyvbjerg, B. (2006). Five misunderstandings about case-study research. *Qual. Inq.* 12 (2), 219–245. doi: 10.1177/1077800405284363

Friesen, N. (2021). "Content in American educational discourse: the missing link(s)," in *Didaktik and Curriculum in Ongoing Dialogue*, eds. E. Krogh, A. Qvortrup, and S. T. Graf (London and New York: Routledge), 65–83.

Gadamer, H.-G. (1960/2010). *Wahrheit und Methode. Grundzüge Einer Philosophischen Hermeneutik*, 7th Edn. Tübingen: Mohr Siebeck.

Gadamer, H.-G. (1960/2006). *Truth and Method*, 2nd revised Edn. New York: Continuum. Translated (1989) by Joel Weinsheimer and Donald Marshall.

Gerring, J., and Cojocaru, L. (2016). Selecting cases for intensive analysis: a diversity of goals and methods. *Sociol. Methods Res.* 45 (3), 392–423. doi: 10.1177/0049124116631692

Grondin, J., and Plant, K. (2003). *The Philosophy of Gadamer*. Durham: Acumen Publishing.

Gustavsson, B. (2014). Bildung and the road from a classical into a global and postcolonial concept. *Confero* 2 (1), 109–131. doi: 10.3384/confero.2001-4562.140604b

Habermas, J. (1978). "Knowledge and human interests," in *Appendix: Knowledge and Human Interests: A General Perspective*. 2nd ed, (Heinemann: London), 301–350. Trans. Shapiro.

Hultengren, E. (1976). *Problemorientering, Projektarbejde og Rapportskrivning. [Problem-orientation, Project Work and Report Writing]*. Aalborg: Institut for Uddannelse og Socialisering, Aalborg Universitetscenter.

Hultengren, E. (1979). *Problemorientering, Projektarbejde og Rapportskrivning. [Problem-orientation, Project Work and Report Writing]*, 2nd Edn. Aalborg: Institut for Uddannelse og Socialisering, Aalborg Universitetscenter.

Hultengren, E. (2014). ""Og det var dengang" – introduktion af problem- og projektbaseret læring på AAU. [That was then: introducing problem- and project-based learning at Aalborg university]," in *Aalborgmodellen – Problembaseret Læring. Aalborg Universitet 40 år [the Aalborg Model – Problem-Based Learning. Aalborg University: 40 Years]*, eds. I. Askehave, A. Bonde, A. Clausen, and P. Plenge (Aalborg: Aalborg Universitet, AAU Kommunikation), 5–9.

IAS-PBL. (2026). *Interdisciplinary student collaboration*. Available online at: https://www.iaspbl.aau.dk/aau-teaching-staff/interdisciplinary-student-collaboration (Accessed August 5, 2026).

Illeris, K. (1974). *Problemorientering og Deltagerstyring: Oplæg til en Alternativ Didaktik. [Problem Orientation and Participant Direction: Proposal of an Alternative Didactic]*. Copenhagen: Munksgaard.

Illeris, K. (1981). *Modkvalificeringens Pædagogik. Problemorientering, Deltagerstyring og Eksemplarisk Indlæring. [The Pedagogy of Counter-Qualification: Problem Orientation, Participant Direction, and Exemplary Learning]*. Copenhagen: Unge Pædagoger.

Illeris, K. (1991). Project education in Denmark. *Int. J. Proj. Manag.* 9 (1), 45–48. doi: 10.1016/0263-7863(91)90055-Z

Jensen, A. A., Stentoft, D., and Ravn, O. (2019). *Interdisciplinarity and Problem-Based Learning in Higher Education: Research and Perspectives from Aalborg University*. Cham: Springer. doi: 10.1007/978-3-030-18842-9

Jæger, K. (2020). Crossing university boundaries: students' experience of PBL as a new epistemic game. *Globe* 9, 29–47. doi: 10.5278/ojs.globe.v9i.4288

Kemp, P. (2011). *Citizen of the World: The Cosmopolitan Ideal for the Twenty-First Century. [Contemporary Studies in Philosophy and the Human Sciences]*. New York: Humanity Books.

Kemp, P. (2015). *Løgnen om Dannelse: Opgør med Halvdannelsen. [The Lie of Bildung: A Reckoning with Half-Education]*. Copenhagen: Tiderne Skifter.

Klein, J. (2009). "The rhetoric of interdisciplinarity. Boundary work in the construction of new knowledge," in *The SAGE Handbook of Rhetorical Studies*, eds. A. A. Lunsford, K. H. Wilson, and R. A. Eberly (Los Angeles, London, New Delhi, Singapore and Washington DC: SAGE), 409–435.

Klindt, M.-P. (2025). Striking a new balance?: courses, project work, and problem-based learning. *J. Probl.-Based Learn. High. Educ.* 13 (1), 310–322. doi: 10.54337/ojs.jpblhe.v13i1.10438

Lindvig, K. E., and Ulriksen, L. (2019). Different, difficult, and local: a review of intedisciplinary teaching activities. *Rev. High. Ed.* 43 (2), 697–725. doi: 10.1353/rhe.2019.0115

Lindvig, K. (2022). "Loud and soft voices of interdisciplinarity in higher education," in *Configurations of Interdisciplinarity in Higher Education*, eds. T. Øland, S. Sauzet, M. L. Ryberg, and K. Lindvig (London: Routledge), 94–118. doi: 10.4324/9781003083245

Lindvig, K. E. (2024). "Weaving interdisciplinarity into disciplinary structures," in *Handbook of Interdisciplinary Teaching and Administration*, ed. R. Szostak (Cheltenham: Edward Elgar Publishing), 436–453.

Lyall, C., Meagher, L., Bandola Gill, J., and Kettle, A. (2015). *Interdisciplinary Provision in Higher Education: Current Context and Future Challenges*. Edinburgh: University of Edinburgh. Available online at: https://www.research.ed.ac.uk/en/publications/interdisciplinary-provision-in-higher-education-current-context-a (Accessed August 5, 2026).

McArthur, J. (2010). Time to look anew: critical pedagogy and disciplines within higher education. *Stud. High. Educ.* 35 (3), 301–315. doi: 10.1080/03075070903062856

Merleau-Ponty, M. (2012). *Phenomenology of Perception*. Abingdon, Oxon: Routledge. Landes, D. A. (translator). doi: 10.4324/9780203720714

Muller, J. (2000). *Reclaiming Knowledge: Social Theory, Curriculum and Education Policy*. London: Routledge.

Muller, J., and Young, M. (2019). Knowledge, power, and powerful knowledge re-visited. *Curric. J.* 30 (2), 196–214. doi: 10.1080/09585176.2019.1570292

Negt, O. (1971). *Soziologische Phantasie und exemplarisches Lernen. Zur Theorie der Arbeiterbildung [Sociological imagination and exemplary learning: on the theory of workers' education]*. Frankfurt a. M.: Europäische Verlagsanstalt.

Negt, O. (1994). "Hvad skal en arbejder vide og forstå for at finde sig tilrette i verden idag? – spørgsmålet om nye samfundsmæssige nøglekvalifikationer. [What must a worker know and understand in order to find their place in the world today? – the question of new societal key qualifications]," in *Arbejde og Subjektivitet – En Antologi om Arbejde, køn og Erfaring. [Work and Subjectivity – An Anthology on Work, Gender and Experience]*, eds. B. S. Nielsen, K. Larsen, H. S. Olesen, and K. Weber (Roskilde: Roskilde Universitetscenter), 43–62.

Nielsen, B. S. (1997). "Det eksemplariske princip. [The principle of exemplarity]," in *Modet til Fremtiden. [Courage for the Future]*, eds. K. Weber, B. S. Nielsen, and H. S. Olesen (Roskilde: Roskilde Universitetsforlag), 269–316.

Ogden, T. H. (1990). *The Matrix of the Mind: Object Relations and the Psychoanalytic Dialogue*. Northvale, N.J.: Jason Aronson, Inc.

Rousseau, J. (1979). *Emile - or, on Education (Book I and II)*. New York: Basic Civitas Books. Translated by A. Bloom.

Servant-Miklos, V. F. C., and Spliid, C. M. (2017). The construction of teaching roles at Aalborg university centre, 1970–1980. *Hist. Educ.* 46 (6), 788–809. doi: 10.1080/0046760X.2017.1360402

Servant-Miklos, V. F. C. (2020). Problem-oriented project work and problem-based learning: "mind the gap!". *Interdiscip. J. Prob.-Based Learn.* 14 (1), 1–17. doi: 10.14434/ijpbl.v14i1.28596

Servant-Miklos, G. (2024). *Pedagogies of Collapse: A Hopeful Education for the End of the World as We Know it*. London: Bloomsbury Academic.

Slothuus, L. (2025). "The climate crisis and interdisciplinary pedagogy," in *Interdisciplinary Pedagogy in Theory and Practice*, eds. L. Slothuus, D. Ashby, and C. Duxbury (London: Routledge), 32–47.

Sørensen, A. (2015). Not work, but alienation and education. *Hegel-Studien* 49, 57–88. Available online at: https://www.jstor.org/stable/10.2307/26595903 (Accessed August 5, 2026).

Sørensen, A. (2021). Educating citizens through Bildung. Didactics and pedagogy in the service of democracy. *J. Educ. Philos. Sociol.* 2 (1), 1–30. doi: 10.29329/jeps.2021.438.1

Thøgersen, U. (2010). Begær som pædagogisk tema – en læsning af rousseaus Émile. [Desire as a pedagogical theme. A Reading of Rousseau's Émile]. *Nord. Stud. Educ.* 30 (1), 49–62. doi: 10.18261/ISSN1891-5949-2010-01-04

Visholm, S. (2004). "Organisationspsykologi og psykodynamisk systemteori. [Management psychology and psychodynamic system theory]," in *Psykodynamisk Organisationspsykologi*, eds. Torben Heinskou, and Steen Visholm (Copenhagen: Hans Reitzels Forlag), 23–48.

Wagenschein, M. (2000). "Teaching to understand: on the concept of the exemplary in teaching," in *Teaching as a Reflective Practice. the German Didaktik Tradition*, eds. I. Westbury, S. Hopmann, and K. Riquarts (London: Routledge), 161–176. doi: 10.4324/9780203357781

Whitehead, J. (1981). Denmark's two university centres: the quest for stability, autonomy, and distinctiveness. *High. Educ.* 10 (1), 89–101. doi: 10.1007/BF00154896

Winnicott, D. W. (1971). *Playing and Reality*. London: Routledge.

Young, M. (2013). Overcoming the crisis in curriculum theory: a knowledge-based approach. *J. Curric. Stud.* 45 (2), 101–118. doi: 10.1080/00220272.2013.764505